# Kubernetes Deployment

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# Agenda

- Kubernetes Working
- Overview – Deployment
- Labels & Selectors
- Equality-based and Set-based Selectors

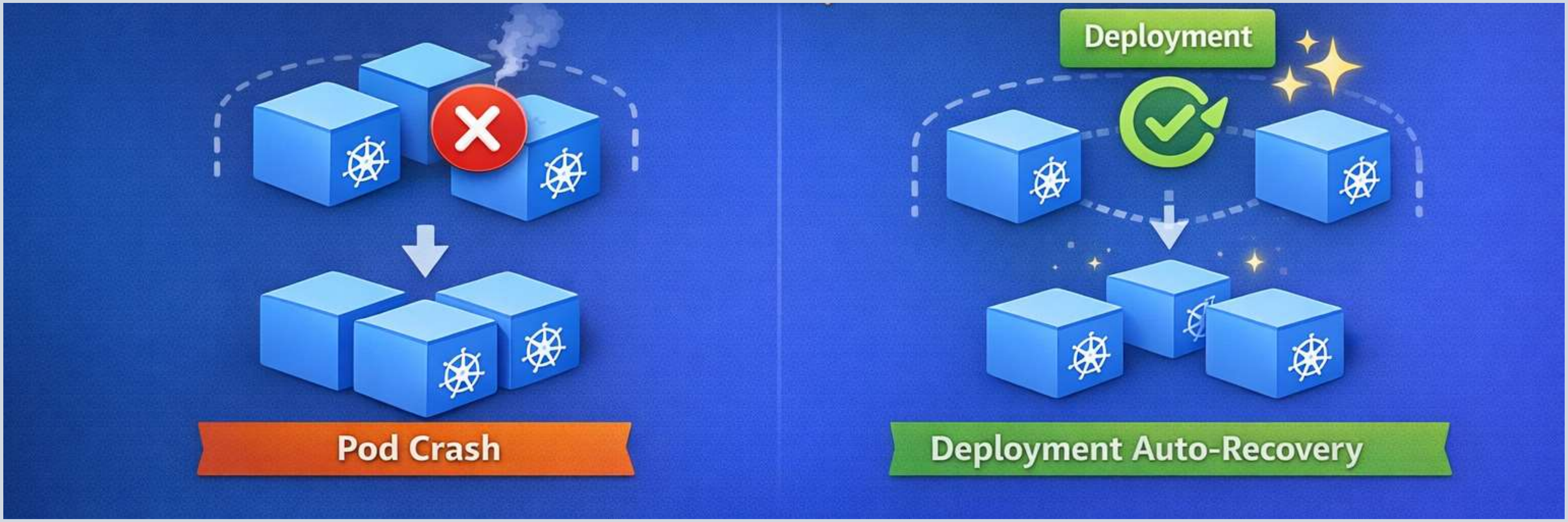
# Why Do We Need Deployments?

**Problem with Pods**

- Pods are ephemeral
- If a Pod crashes → manual recreation
- No rollout or rollback
- No scaling automation

**What Deployments Solve**

- High availability
- Zero-downtime updates
- Self-healing
- Declarative management




# Why Do We Need Deployments?

# What Is a Deployment?

**Definition**

- A Deployment is a **higher-level controller**
- Manages **ReplicaSets**
- ReplicaSets manage **Pods**

**In Simple Terms**

> "Deployment = Desired state manager for your application"

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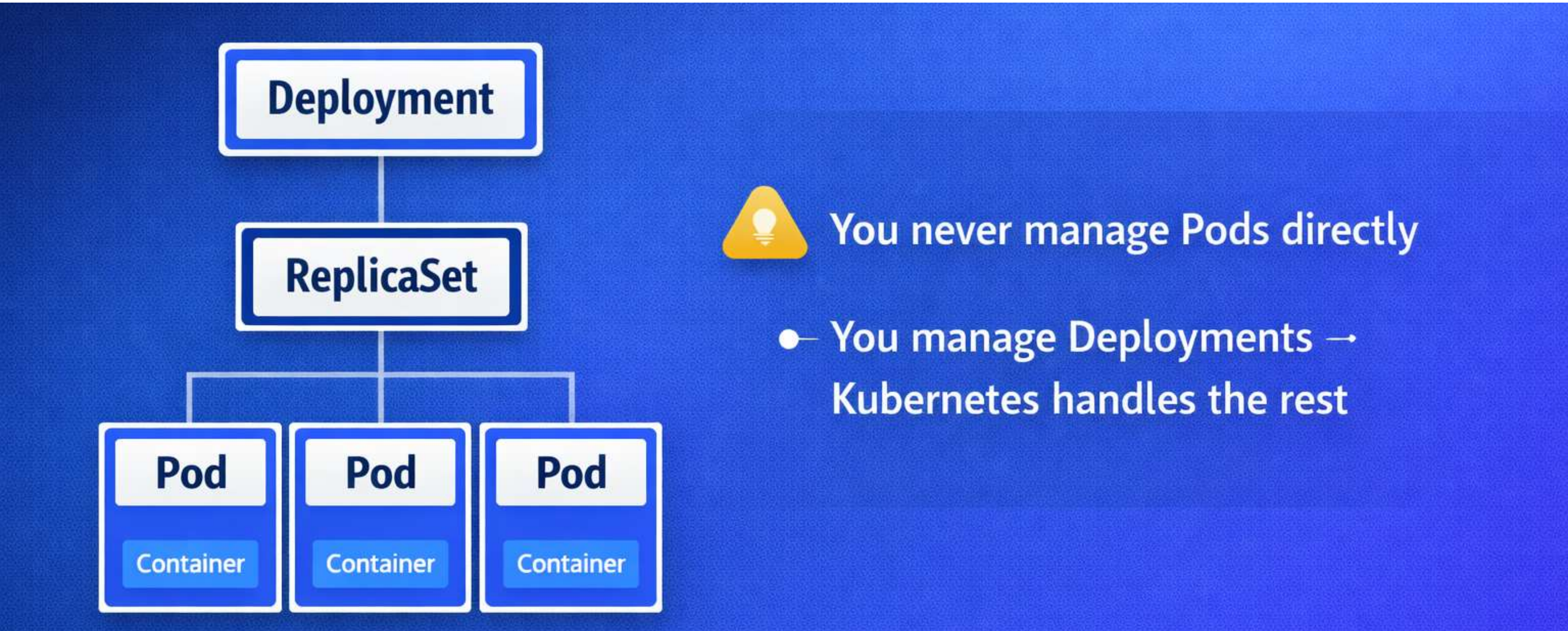



# Deployment Architecture

# Deployment vs Pod vs ReplicaSet

| Feature | Pod | ReplicaSet | Deployment |
|---|---|---|---|
| Self-healing | ❌ | ✅ | ✅ |
| Rolling updates | ❌ | ❌ | ✅ |
| Rollback | ❌ | ❌ | ✅ |
| Scaling | ❌ | ✅ | ✅ |

# De-coding the YAML file

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
```

```yaml
template:
   metadata:
     labels:
       app: nginx
spec:
     containers:
     - name: nginx
       image: nginx:1.25
       ports:
       - containerPort: 80
```


# Basic Deployment YAML – Structure Overview

# Understanding replicas

- Number of Pod copies
- Ensures **desired state**
- If 1 Pod dies → new Pod created automatically

Example

```
replicas: 3
```

# Selector & Labels – The Glue

**Labels**

- Key-value metadata
- Used for grouping resources

**Selector**

- Deployment uses selector to find Pods it manages

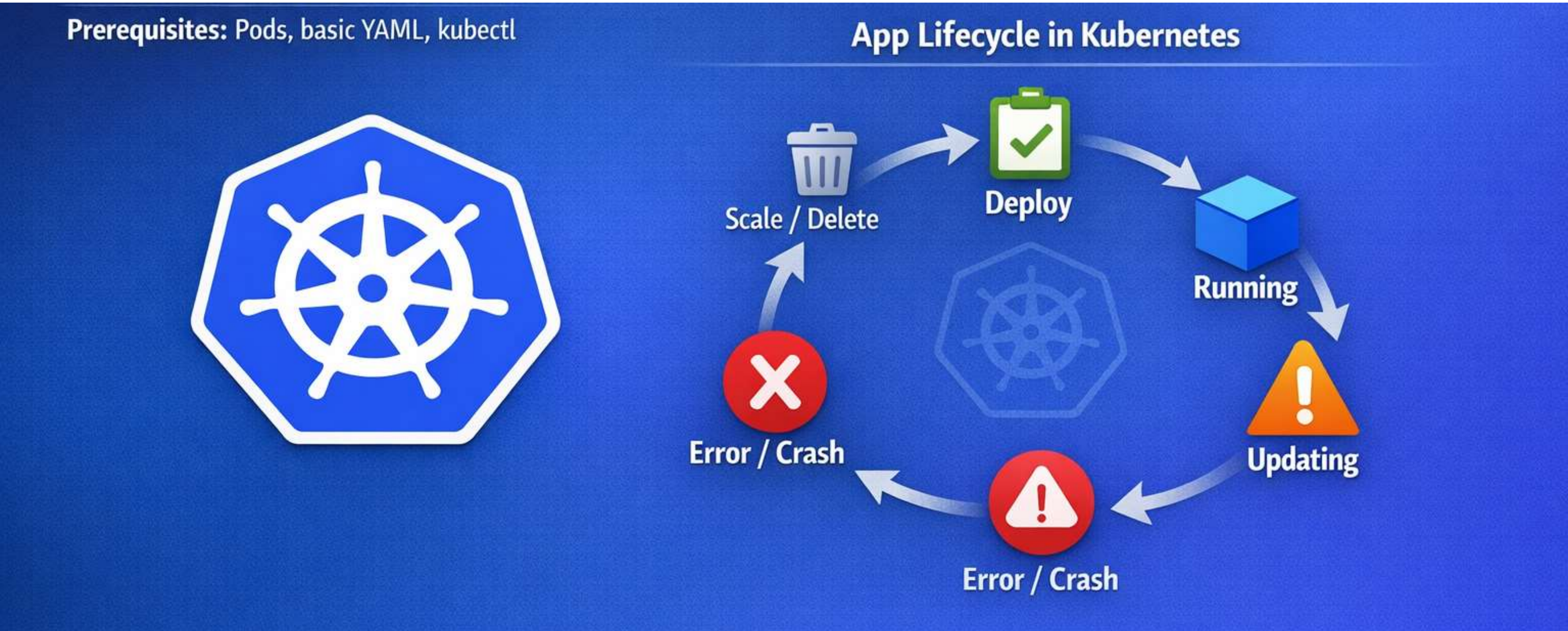
⚠️ **Important Rule**

- *selector.matchLabels* **must match** Pod template labels

# Pod Template – Heart of Deployment

```
template:
  metadata:
    labels:
      app: nginx
  spec:
    containers:
    - name: nginx
      image: nginx
```

- Any change here → new ReplicaSet created
- Triggers rollout




# Kubernetes Deployment – App Lifecycle

# Creating a Deployment (kubectl)

```
kubectl apply -f deployment.yaml
```

**Verify**

```
kubectl get deployments
kubectl get rs
kubectl get pods
```



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# Scaling a Deployment

**Manual Scaling**

```
kubectl scale deployment nginx-deployment --replicas=5
```

**YAML-based Scaling**

```
replicas: 5
```



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# Rolling Updates – Core Feature

**What Happens During Update**

- New ReplicaSet created
- Pods updated gradually
- Old Pods terminated slowly

**Example**

```
kubectl set image deployment/nginx-deployment nginx=nginx:1.26
```

# Rollback a Deployment

**Check History**

```
kubectl rollout history deployment nginx-deployment
```

**Rollback**

```
kubectl rollout undo deployment nginx-deployment
```

# Deployment Status & Monitoring

```
kubectl rollout status deployment nginx-deployment
kubectl describe deployment nginx-deployment
```

**What You See**

- Progress
- Errors
- Events

# Pausing & Resuming Deployments

**Pause**

```
kubectl rollout pause deployment nginx-deployment
```

**Resume**

```
kubectl rollout resume deployment nginx-deployment
```

# Handling Failed Deployments

**Common Causes**

- Image pull error
- App crash
- Readiness probe failure

**How Kubernetes Reacts**

- Stops rollout
- Keeps old Pods running

# Rolling Update Strategy

```
strategy:
  type: RollingUpdate
  rollingUpdate:
    maxUnavailable: 1
    maxSurge: 1
```

**Meaning**

- maxUnavailable: Pods allowed to go down
- maxSurge: Extra Pods allowed temporarily

# Provide resource config

```yaml
resources:
  requests:
    cpu: "100m"
    memory: "128Mi"
  limits:
    cpu: "500m"
    memory: "256Mi"
```

**Why Important**

- Prevents noisy neighbor problem
- Enables autoscaling

# Common Beginner Mistakes

- Editing Pods directly
- Label mismatch
- Forgetting resource limits
- Using latest image tag
- No probes configured